Uniform Feature Routing in Controlled Top-1 Mixture-of-Experts

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Abstract

This paper studies a minimal controlled question behind uniform feature routing in top-1 mixture-of-experts (MoE): when do uniformly occurring features route to uniformly used experts without an explicit balancing rule? We show that the answer is geometric rather than count-based. If route-relevant token directions are uniform on the sphere and expert prototypes induce equal-measure spherical cells, then unconstrained top-1 routing has uniform expected expert usage; a regular simplex gives a constructive case. This condition separates natural geometric balance from two commonly conflated effects: finite-token sampling noise and capacity-constrained assignment. We also clarify the role of spectral evidence. A flat hidden covariance spectrum is a useful diagnostic but not a usage-balance theorem, since covariance identity need not imply directional uniformity. Conversely, in a controlled elliptical family, whitening recovers the spherical model. Expert principal singular directions, including those used by MPI-style routers, are therefore candidate centers rather than balance guarantees. Synthetic experiments validate these controlled claims and support no stronger conclusion about natural language semantics or training-time preservation.

1 Introduction

Sparse mixture-of-experts (MoE) layers increase parameter capacity by routing each token to a small subset of experts (Shazeer et al., 2017; Fedus et al., 2022). In a linear top-1 router, a token hidden state is assigned to the expert whose router row has the largest inner product with it. Therefore, when we ask whether uniformly occurring features can be routed to uniformly used experts, the immediate mathematical object is not the language task, but the geometry seen by this inner product.

The controlled setting is deliberately simpler than real language modeling. Consider many independent feature families with equal frequency and no semantic relation across families. Even in this simplified regime, several notions of "uniform" must be separated. Feature labels can be frequency-balanced while the hidden states are not uniformly distributed in high-dimensional space. The hidden covariance can have no dominant eigendirection while the direction distribution is still not spherical. Expert usage counts can be forced to be exactly equal by a capacity rule, but this does not explain why top-1 routing would balance naturally. Conversely, router rows can align with expert singular directions while still producing imbalanced usage when those directions cluster.

We therefore restrict the paper to the static controlled case. We do not claim that a short-context next-token-prediction model naturally learns uniform hidden states; such states may be strongly anisotropic because of local prediction signals and residual common components. Instead, we ask: under what representation-geometric assumptions is uniform expert usage a consequence of unconstrained top-1 routing itself?

Main message. Uniform top-1 expert usage is a two-sided sufficient-condition result. On the token side, the route-relevant feature directions must be close to a spherical uniform distribution. On the expert side, the router rows or expert-derived prototypes must cut that

sphere into equal-measure cells. If either side fails, usage can be imbalanced even when the other side is well behaved.

Contributions.

1. We formulate the controlled uniform feature routing problem and separate feature-frequency uniformity, hidden spectral uniformity, directional uniformity, and expert-usage uniformity.
2. We prove a d-dimensional sufficient condition for natural uniform top-1 expert usage and give a regular-simplex corollary.
3. We decompose finite-token observed imbalance into a fixed prototype-cell term and a sampling-noise term.
4. We show that covariance spectral uniformity is only a second-order diagnostic by giving an isotropic-covariance counterexample with nonuniform routing.
5. We connect expert principal singular directions and MPI-style routing to our story as center construction, not as a balance theorem.
6. We provide synthetic experiments matching each theoretical claim and keep balanced assignment only as a non-geometric control.

2 Problem Setup

2.1 Top-1 routing and expert usage

Let $h_i \in \mathbb{R}^d$ be the route-relevant representation of example i. The top-1 router has one prototype $u_e \in \mathbb{R}^d$ for expert $e \in \{1, \dots, E\}$. It computes

$$s_e(h_i) = \langle h_i, u_e \rangle, \qquad a(h_i) = \arg\max_e s_e(h_i). \tag{1}$$

The empirical expert usage is

$$\widehat{p}_e = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{a(h_i) = e\}. \tag{2}$$

The count-level target is $\widehat{p}_e \approx 1/E$ for all experts. This paper asks when that target arises from the top-1 geometry itself, not from an external capacity constraint.

2.2 Four notions of uniformity

Definition 1 (Feature-frequency uniformity). Feature labels $f \in \{1, \dots, F\}$ are frequency-uniform if $\mathbb{P}(f) = 1/F$ for every f.

Definition 2 (Directional uniformity). After centering and normalizing route-relevant states,

$$z_i = \frac{h_i - \mu}{\|h_i - \mu\|_2}, \tag{3}$$

the token-side representation is directionally uniform if z_i is approximately uniform on $\mathbb{S}^{d-1}$.

Definition 3 (Hidden spectral uniformity). Let $\bar{h}_i = h_i - \mu$ and

$$C = \frac{1}{N} \sum_i \bar{h}_i \bar{h}_i^\top. \tag{4}$$

The representation is spectrally uniform when C has no dominant eigendirection, ideally $C = \sigma^2 I$. We use condition number, spectral entropy, and effective rank as diagnostics.

Definition 4 (Expert-usage uniformity). The router is usage-uniform when $\widehat{p}_e \approx 1/E$ for all experts.

The order matters. Frequency uniformity is a data property. Directional uniformity is the token-side condition directly seen by the inner-product router. Hidden spectral uniformity is a second-order diagnostic that can help detect or repair anisotropy, but it is not the same as directional uniformity. Usage uniformity is the router outcome.

3 Theory

3.1 A sufficient condition for natural top-1 balance

Normalize token features and expert prototypes to the sphere: $z \in \mathbb{S}^{d-1}$ and $u_e \in \mathbb{S}^{d-1}$. Prototype u_e owns the spherical top-1 cell

$$V_e = \{z \in \mathbb{S}^{d-1} : \langle z, u_e \rangle \geq \langle z, u_j \rangle,\ \forall j\}. \tag{5}$$

Tie boundaries are measure-zero under the continuous spherical uniform distribution, so the tie-breaking rule does not affect the result below.

Proposition 1 (Equal-measure cells are sufficient). Assume $z \sim \mathrm{Unif}(\mathbb{S}^{d-1})$ and the cells $V_1, \ldots, V_E$ form an equal-measure partition of the sphere up to measure-zero boundaries. Then

$$\mathbb{P}(a(z) = e) = \frac{1}{E}, \qquad \forall e, \tag{6}$$

and for independent examples, $\mathbb{E}[\widehat{p}_e] = 1/E$.

Proof. The event $a(z) = e$ is exactly $z \in V_e$. The uniform distribution on the sphere assigns probability according to spherical measure. Since the E cells cover the sphere up to measure-zero boundaries and have equal measure, each cell has probability mass $1/E$.

Corollary 1 (Regular-simplex construction). If $E \leq d+1$ and $u_1, \ldots, u_E$ are the vertices of a regular simplex centered at the origin, then the induced top-1 cells have equal spherical measure. Therefore uniform spherical features have uniform expected top-1 expert usage.

Reason. A regular simplex is invariant under all permutations of its vertices. These permutations are realized by orthogonal transformations of the span of the simplex and can be extended by the identity on the orthogonal complement in $\mathbb{R}^d$. Thus every cell can be mapped to every other cell by a measure-preserving orthogonal map. The cells are congruent and have equal spherical measure.

3.2 Finite tokens do not fix prototype-cell imbalance

For any fixed prototype geometry, define the population cell mass $p_e = \mathbb{P}(z \in V_e)$. Then

$$\widehat{p}_e - \frac{1}{E} = \left(p_e - \frac{1}{E}\right) + (\widehat{p}_e - p_e)\,. \tag{7}$$

The first term is the prototype-cell geometry error. The second term is finite-token sampling noise. Increasing N reduces the second term but does not change the first.

In the two-dimensional circle model, this decomposition is explicit. If E random centroids are sampled uniformly on the circle and the sorted circular gaps are $G_1, \ldots, G_E$, then

$$(G_1, \ldots, G_E) \sim \mathrm{Dirichlet}(1, \ldots, 1), \qquad p_j = \frac{G_{j-1} + G_j}{2}. \tag{8}$$

Thus $\mathbb{E}[p_j] = 1/E$, but a single initialization is not balanced. More routed tokens estimate these unequal p_j more accurately; they do not make the p_j equal.

3.3 Spectral uniformity is only a diagnostic

A flat covariance spectrum is useful, but it is not equivalent to the spherical condition in Proposition 1.

Proposition 2 (Isotropic covariance does not imply directional uniformity). There exists a zero-mean distribution in $\mathbb{R}^2$ with covariance $I/2$ whose top-1 load under equal-measure simplex prototypes is not uniform.

Proof. Let the token direction be uniform over the four axis points $\{(1,0),(0,1),(-1,0),(0,-1)\}$. The mean is zero and the covariance is $I/2$. Now take three unit prototypes at angles 20°, 140°, and 260°. These are regular-simplex vertices on the circle, so under the continuous uniform circle distribution their cells have equal measure 1/3. For the four-axis distribution, the points at angles 0°, 90°, 180°, and 270° route respectively to the first, second, second, and third prototypes. The induced load is therefore $(1/4, 1/2, 1/4)$, not $(1/3, 1/3, 1/3)$. Hence covariance identity alone does not certify directional uniformity or load uniformity.

This counterexample is the formal reason we treat hidden spectral uniformity as a second-order diagnostic. It can support the spherical model in controlled families, but it cannot replace a directional or cell-mass analysis.

3.4 Controlled whitening for an elliptical toy family

A related question is what happens when hidden states contain a large singular direction. We model this effect by applying an elliptical distortion to a spherical feature distribution.

Proposition 3 (Population whitening recovers spherical directions in the elliptical toy model). Let $z \sim \mathrm{Unif}(\mathbb{S}^{d-1})$ and $h = \mu + Az$, where A is invertible. The population covariance is $C_h = AA^\top/d$. If $x = C_h^{-1/2}(h-\mu)$, then $x = \sqrt{d}\,Qz$ for an orthogonal matrix Q. Therefore $x/\|x\|_2$ is again uniform on $\mathbb{S}^{d-1}$.

Proof. Since $\mathbb{E}[zz^\top] = I/d$, $C_h = AA^\top/d$. Therefore

$$C_h^{-1/2}(h-\mu) = \sqrt{d}\,(AA^\top)^{-1/2}Az. \tag{9}$$

By the polar decomposition, $(AA^\top)^{-1/2}A$ is orthogonal. The spherical uniform distribution is invariant under orthogonal transformations, so the normalized whitened direction is spherical uniform.

Boundary. This proposition justifies the elliptical toy experiment. It does not say that arbitrary trained hidden states are elliptical, nor that empirical covariance identity implies spherical direction uniformity.

4 Spectral Expert Prototypes Identify Centers, Not Balanced Cells

The theory above assumes expert prototypes are given. Expert matrices can provide candidate prototypes through singular directions. Let W_e be an input-side expert matrix. Define its spectral prototype by

$$v_e \in \arg\max_{\|v\|_2=1} \|v^\top W_e\|_2^2. \tag{10}$$

Equivalently, v_e is a top eigenvector of $W_eW_e^\top$. MPI-style routers make this connection explicit by applying a power-then-retract update (Wu et al., 2026):

$$\widehat{r}_e = r_eW_eW_e^\top, \qquad r_e' = C_r\frac{\widehat{r}_e}{\|\widehat{r}_e\|_2}. \tag{11}$$

This update can align a router row with an expert's principal direction. But alignment and uniform usage answer different questions. Alignment says a row is a good expert proxy. Uniform usage requires the set of recovered proxies to produce equal-mass cells for the token direction distribution. If the expert spectral directions are clustered around a common component, better alignment can preserve or amplify imbalance rather than remove it.

5 Capacity Assignment: A Non-Geometric Control

If the only target is exact usage counts in one finite batch, a capacity rule can force them. Given scores $S_{ie} = \langle z_i, u_e \rangle$, balanced assignment solves

$$\max_{A} \quad \sum_{i=1}^{N} \sum_{e=1}^{E} A_{ie} S_{ie}, \tag{12}$$

$$\text{s.t.} \quad \sum_{e} A_{ie} = 1, \qquad A_{ie} \in \{0, 1\}, \tag{13}$$

$$\sum_{i} A_{ie} \in \left\{ \left\lfloor \frac{N}{E} \right\rfloor, \left\lceil \frac{N}{E} \right\rceil \right\}. \tag{14}$$

This is important as a control and as a strict finite-token construction. It is not evidence that the representation is uniform, that features are meaningful, or that expert specialization has utility.

6 Experiments

6.1 Setup and metrics

All experiments are synthetic and static. They test exactly the controlled claims above. The main metric is LoadCV, the coefficient of variation of empirical expert usage. It is zero under exact count balance and increases as the usage distribution becomes more uneven:

$$\text{LoadCV} = \frac{\text{std}(\widehat{p}_1, \ldots, \widehat{p}_E)}{\text{mean}(\widehat{p}_1, \ldots, \widehat{p}_E)}. \tag{15}$$

For spectral geometry, we report covariance condition number and effective rank. For spectral expert prototypes, we report the normalized alignment

$$\lambda_e = \frac{\|r_e^\top W_e\|_2}{\|r_e\|_2 \|W_e\|_2}. \tag{16}$$

6.2 E1: prototype geometry controls uniform-sphere usage

E1 samples $N = 100000$ examples from $\mathbb{S}^{63}$ and routes them to $E = 8$ experts. Symmetric prototypes are regular-simplex rows. Random prototypes are independent unit vectors. Clustered prototypes are perturbations of one shared direction.

Table 1: d-dimensional uniform-sphere routing with $d = 64$ and $E = 8$.

Prototype geometry	LoadCV ↓	Max/min load ↓	Max abs. error ↓
Symmetric top-1	0.0067	1.021	0.00145
Random top-1	0.0657	1.227	0.01451
Clustered top-1	0.1349	1.574	0.03580

Table 1 and Figure 1 support the sufficient-condition story: well-spread cells give near-uniform usage, whereas clustered cells do not.

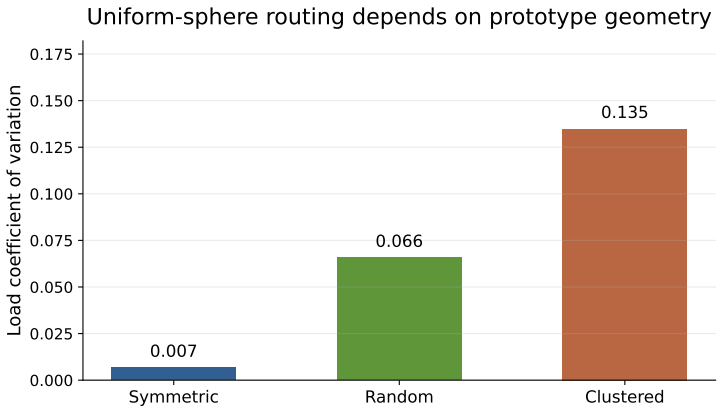


Figure 1: Uniform spherical features do not alone determine expert usage. Prototype geometry controls load variation.

6.3 E2: random prototype imbalance is a geometry floor

E2 estimates high-dimensional random-prototype cell masses by Monte Carlo reference sampling. For a fixed random prototype realization, we compare geometry error, sampling error, and total usage error as routed batch size increases.

Table 2: High-dimensional random-prototype floor for $d = 64$. Sampling error shrinks with routed batch size; the geometry floor is fixed for the same prototype realization.

Experts	Examples	Geometry error ↓	Sampling error ↓	Total error ↓
4	10^3	0.093	0.074	0.096
4	$5 \cdot 10^4$	0.093	0.014	0.097
8	10^3	0.093	0.160	0.182
8	$5 \cdot 10^4$	0.093	0.031	0.109
16	10^3	0.185	0.237	0.234
16	$5 \cdot 10^4$	0.185	0.044	0.194

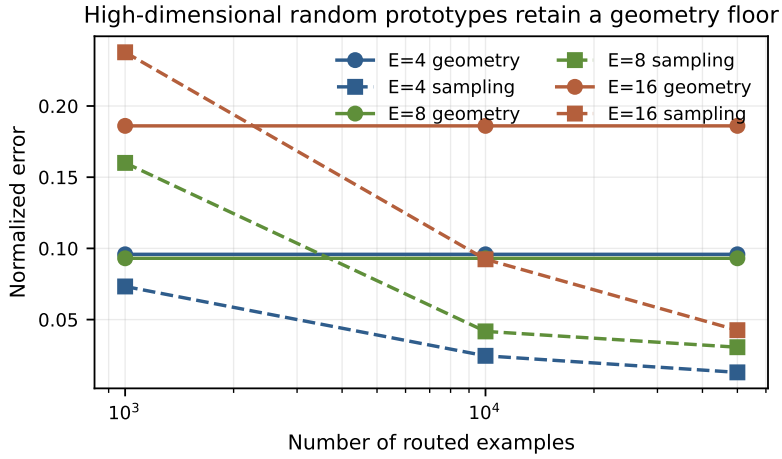


Figure 2: Increasing routed examples reduces sampling noise but does not remove the single-realization random-prototype geometry floor.

This directly matches the finite-token decomposition: more examples reduce $\widehat{p}_e - p_e$, not $p_e - 1/E$.

6.4 E3: spectral prototypes are centers, not balance

E3 constructs expert matrices with prescribed input-side top singular directions,

$$W_e = \sigma_1 v_e a_e^\top + \sigma_0 \Xi_e. \tag{17}$$

The prescribed directions are symmetric, random, or clustered. We compare exact SVD, one-step MPI, five-step MPI, and random rows.

Table 3: Spectral alignment recovers expert directions, but usage balance still depends on prototype spread.

Prototype geometry	Router	Alignment ↑	LoadCV ↓	Max/min load ↓
Symmetric	Exact SVD	1.000	0.009	1.030
Symmetric	MPI-1	0.975	0.025	1.094
Clustered	Exact SVD	1.000	0.161	1.631
Clustered	MPI-1	0.977	0.195	1.788
Random rows	No expert alignment	0.098	0.074	1.281

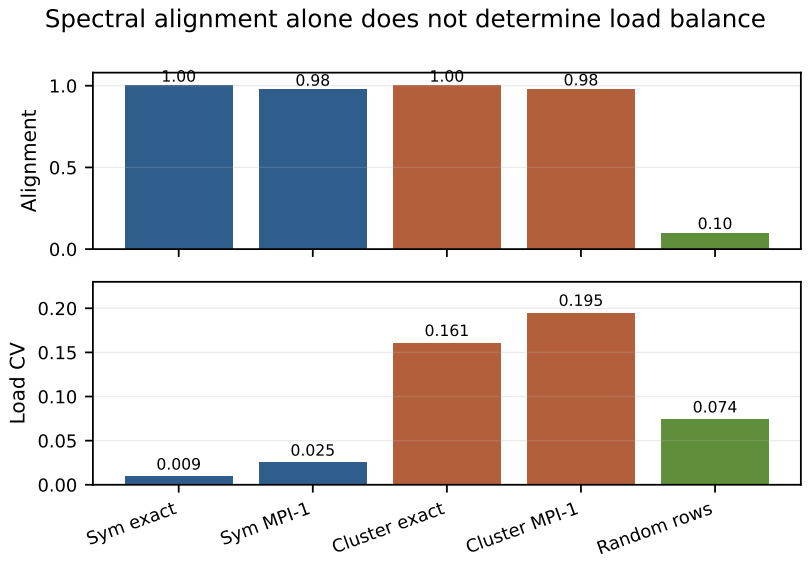


Figure 3: High spectral alignment does not imply balanced expert usage.

The result clarifies the role of spectral router construction. Expert singular directions can be meaningful centers, but a center-construction rule is not a load-balance theorem.

6.5 E4: capacity assignment is only a control

E4 tests the strict finite-batch alternative. With $N = 2048, d = 64, E = 8$, both greedy capacity-feasible assignment and exact balanced assignment over expanded expert slots enforce zero LoadCV.

Table 4: Capacity assignment as a non-geometric control. Exact balanced assignment reaches zero LoadCV with lower score drop than greedy assignment, but it changes the assignment rule rather than the representation geometry.

Prototype geometry	Top-1 LoadCV ↓	Greedy score drop ↓	Exact balanced score drop ↓
Symmetric	0.054	0.00091	0.00012
Random	0.122	0.00416	0.00056
Clustered	0.128	0.00379	0.00047

We keep E4 in the main paper because it prevents a common misreading: uniform expert usage can be forced, but forced usage is not evidence for uniform feature geometry.

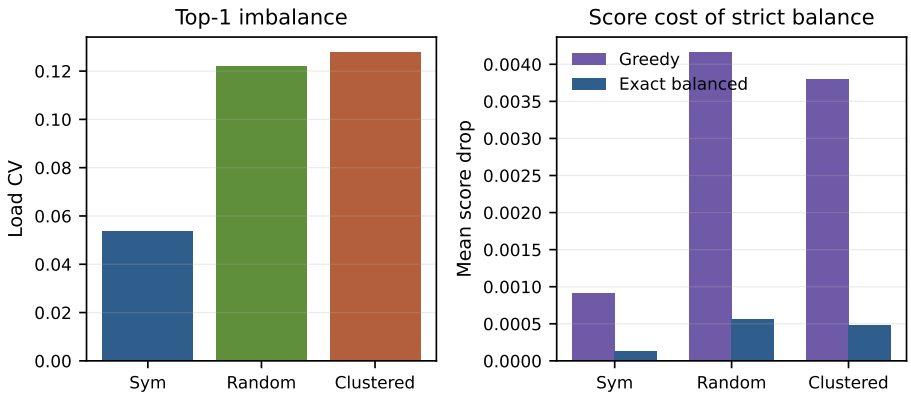


Figure 4: Balanced assignment exposes the cost of forcing the count metric. It is a control for count equality, not the representation mechanism studied here.

6.6 E5: controlled spectral uniformization repairs an elliptical toy

E5 tests the whitening proposition in a controlled family. Starting from $z \sim \mathrm{Unif}(\mathbb{S}^{63})$, we form anisotropic features $h = Dz$ with $D = \mathrm{diag}(\alpha, 1, \ldots, 1)$. We compare raw features, centered-normalized features, partial spectral shrinkage with $\beta = 0.5$, and whitening.

Table 5: Representation-geometric repair for $\alpha = 20$. Whitening makes the toy hidden spectrum nearly uniform and restores the routing diagnostic.

Representation	Condition no. ↓	Effective rank ↑	LoadCV diagnostic ↓
Raw	419.52	2.61	1.053
Centered + normalized	114.13	8.95	1.053
Partial shrinkage	14.86	47.02	0.781
Whitened + normalized	1.00	64.00	0.005

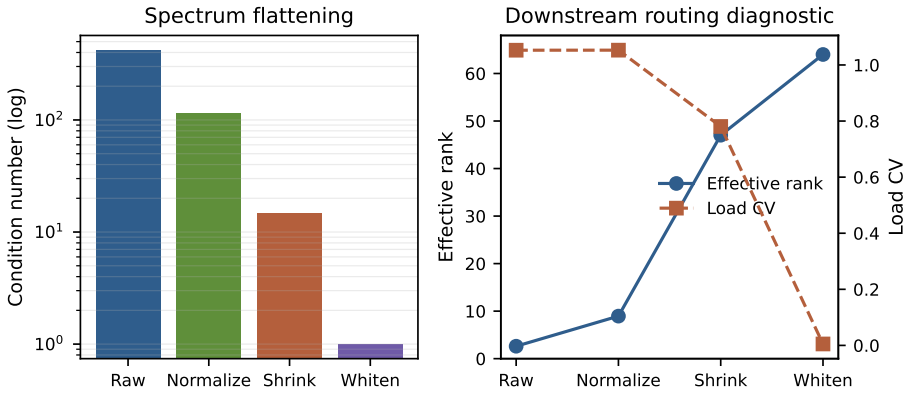


Figure 5: In the elliptical toy setting, spectral flattening moves the representation back toward the spherical routing model.

E5 should be read with the theorem boundary in mind. Whitening is successful here because the data are generated by an elliptical distortion of a spherical source. This is not a claim that arbitrary next-token hidden states are elliptical or naturally whitened.

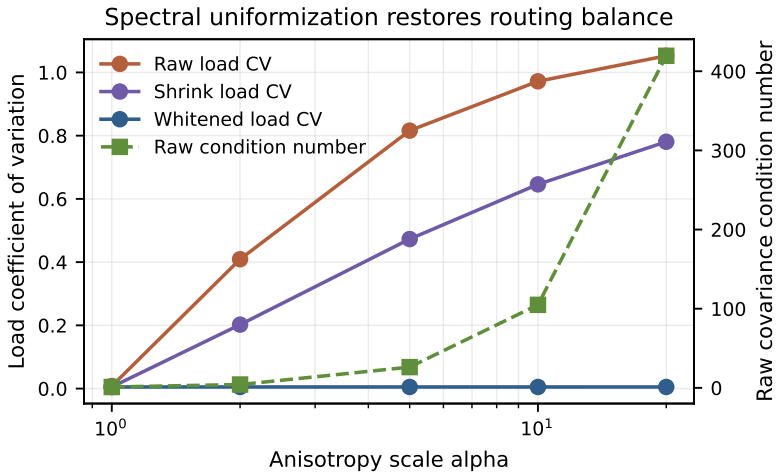


Figure 6: Across anisotropy strengths, spectral shrinkage partially improves the routing diagnostic, while whitening returns the toy representation to the uniform-spectrum regime.

7 Scope and Limitations

All positive statements in this paper are controlled sufficient constructions. The paper does not prove a necessary-and-sufficient theorem for all routers or all hidden-state distributions. It also does not claim that hidden spectral uniformity alone implies load uniformity; Proposition 3 explicitly shows that it does not. Spectral flattening is useful in E5 because the toy data satisfy a strong elliptical assumption.

Training dynamics are outside the main claim. Static uniform assignment and training-time preservation are different questions. During optimization, token features, router rows, and expert matrices can move. Preservation then depends on margin lower tails and drift budgets, not only on step-zero geometry. The short-sequence next-token setting is therefore a later nonuniform-spectrum and margin-preservation problem, not the main evidence for this paper.

8 Conclusion

Uniform expert usage under unconstrained linear top-1 routing is a controlled representation-geometric phenomenon. Equal feature frequency is not enough. Flat covariance is not enough. Expert spectral alignment is not enough. The sufficient condition is two-sided: token directions must match the spherical model, and expert prototypes must induce equal-measure cells. Balanced assignment can force uniform counts, but that is a different mechanism. This paper closes the clean uniform-feature case and identifies the next controlled case: nonuniform hidden spectra and training-time preservation.

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